

# X-RAY DIFFRACTION

## PURPOSE OF EXPERIMENT:

- To investigate the production, diffraction and absorption of x-rays.
- To use the known atomic spacing of a crystal to identify the wavelength of X-Rays.
- To use the wavelength of the X-Rays to find the atomic spacing of a different crystal and the atomic spacing of crystals in different orientations.

APPARATUS: You will be using a “Tel-X-O meter” student x-ray machine – see figure below. This compact unit contains an x-ray tube with a copper anode, a central post for mounting a crystal – NaCl or LiF - a rotatable carriage arm, a Geiger–Müller tube, a “stepper motor/computer interface” and a computer running TEL-X-Driver software. Other auxiliary equipment including collimators and filters may be used. **Note: Please do not touch the crystals** they are quite fragile.

HOW TO USE THE “Tel-X-O meter”: The Geiger–Müller tube, is mounted at the end of the carriage arm and a NaCl (yellow tipped) or LiF (blue tipped) crystal is mounted in the post that defines the precise center of the arms rotation. X-Rays are emitted from the large clear domed object at the back of the machine. Before you begin using the machine, the TA will:

- a. Teach you how to open, close and operate the x-ray machine.
- b. Rotate the carriage arm from  $\sim 20^\circ$  deg through  $\sim 90^\circ$  to demonstrate the  $\Theta/2\Theta$  configuration. The angular position of the carriage arm - relative to the x-ray beam - is always twice the angular position of the crystal. E.g. if the carriage arm is at  $40^\circ$ , the crystal will be at  $20^\circ$ .
- c. Demonstrate how to calibrate the location of the carriage arm in the TEL-X-Driver software.
- d. Go over the safety features of the instrument and point out precautions including:
  - I. Please do not touch the crystals, they are fragile and precisely aligned.
  - II. Please keep the tube current between 65 & 80  $\mu\text{A}$ . **The tube may be damaged if the current is greater than 80  $\mu\text{A}$  for an extended time.**

## PROCEDURE:

There are four x-ray machines setup in the lab labeled A), B), C) & D). Each machine is configured for a different experiment from the “TEL508 X-Ray Machine” manual – pertinent sections are included at the end of this document. You are expected to perform the assigned experiments on all four of the setups:

- A) NaCl @  $0^\circ$ : do Exp #14
- B) LiF @  $0^\circ$ : do Exp #14
- C) NaCl @  $45^\circ$ : do Exp #14 refer to Exp #25
- D) LiF @  $45^\circ$ : do Exp #14 refer to Exp #25

For the complete manual see “Reference and Resources” on the wiki. Please note: the manual was written before the addition of the “stepper motor/computer interface” etc. so the data collection and graphing instructions may not be pertinent.

- 1) Select one of the setups and perform the associated experiment. When done, move to a different setup and perform the assigned experiment. Continue in this fashion until you have worked with all four of the X-Ray machines
  - a. For Setups A&B – Use the TEL-X-Driver software to measure the spectrum of the copper anode x-ray tube via Experiment #14 in the “TEL508 X-Ray Machine” manual. The software will measure the counting rate as a function of the position ( $2\theta$ ) of the carriage arm. Check the calibration of the carriage arm. Do a wide scan using 1 second readings at  $0.5^\circ$  intervals to identify the peaks. Around each peak, scan a narrow angular range with a longer counting time and a smaller step size. Repeat measurements to get more accuracy.
    - I. How can you estimate the accuracy?
    - II. What should the uncertainty be theoretically?
  - b. For Setups C&D - Use the TEL-X-Driver software to explore the crystal lattice. Follow the instructions for Setups A&B (Experiment #14) but refer to the last half of Experiment #25 – around D25.10 - for clarification on the experimental setup and tips on data analysis. The software will measure the counting rate as a function of the position ( $2\theta$ ) of the carriage arm. Check the calibration of the carriage arm. Do a wide scan using 1 second readings at  $0.5^\circ$  intervals to identify the peaks. Around each peak, scan a narrow angular range with a longer counting time and a smaller step size. Repeat measurements to get more accuracy.
    - I. How can you estimate the accuracy?
    - II. What should the uncertainty be theoretically?

Please Remember:

- Keep the tube current between  $65 \text{ \& } 80 \mu\text{A}$  . Check it frequently.
- The peaks in the x-ray spectrum are due to the  $K_\alpha$  and  $K_\beta$  lines.
- The interlayer distance  $d$  in NaCl is  $2.82 \text{ \AA}$ .
- The Bragg condition is  $2d \sin \theta = n\lambda$  .

#### Data Analysis:

1. Use your data to complete the tables etc. described in Experiments #14 and #25 in the “TEL508 X-Ray Machine” manual.
2. Use the NaCl data to find the wavelengths of  $K_\alpha$  and  $K_\beta$  emission.
3. Use the wavelengths of  $K_\alpha$  and  $K_\beta$  to find the lattice constant of LiF.
4. Use the wavelengths of  $K_\alpha$  and  $K_\beta$  to find the lattice constant of the NaCl & LiF crystals in setups C & D.
5. Why are they different than the lattice constant in A& B? Explain.

#### Discussion of Results:

1. Be sure to include a physical interpretation of the data, and a critical evaluation of the data's accuracy.
2. Explain the origin of the main features observed in the x-ray spectrum.
3. Determine the photon energy of the  $K_\alpha$  and  $K_\beta$  lines, estimate the error and compare with the Bohr model predictions.
4. Why don't the  $K_\alpha$  and  $K_\beta$  lines have equally intensity?

# D14 – WAVELENGTH MEASUREMENT: BRAGG METHOD (1½ HOURS)

Sir Lawrence Bragg presumed that the atoms of a crystal such as Sodium Chloride were arranged in a cubic and regular three-dimensional pattern.

The mass of a molecule of NaCl is  $M/N$  Kg, where  $M$  is the molecular weight ( $58.46 \times 10^{-3}$  kg per mole) and  $N$  is Avogadro's number ( $6.02 \times 10^{23}$  molecules per mole).

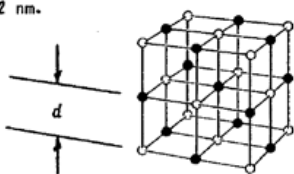
The number of molecules per unit volume is  $\rho \frac{N}{M}$  molecules per cubic metre, where  $\rho$  is the density ( $2.16 \times 10^3$  kg m<sup>-3</sup>).

Since NaCl is diatomic the number of atoms per unit volume is  $2\rho N/M$  atoms per cubic metre.

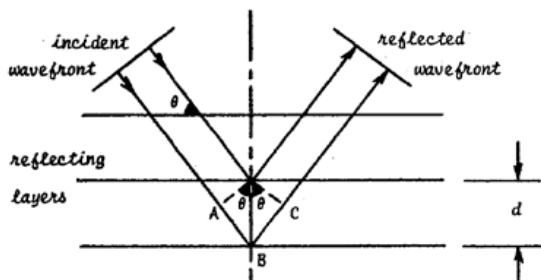
The distance therefore between adjacent atoms,  $d$  in the lattice is derived from the equation

$$d^3 = \frac{1}{2\rho N/M} \quad \text{or} \quad d = \sqrt[3]{M/2\rho N}$$

and for NaCl,  $d = 0.282$  nm.



The first condition, for Bragg "reflection" is that the angle of incidence  $\theta$  equals the angle of reflection – this is as for optical reflection and infers that any detector of the reflected rays must move through an angle  $2\theta$ , the 2:1 spectrometer relationship.



The second condition is that reflections from several layers must combine constructively :-

$$n\lambda = AB + BC = 2d \sin \theta$$

|         |          |            |            |
|---------|----------|------------|------------|
| KIT 582 | 30/20 kV | 50 $\mu$ A | NORMAL LAB |
|---------|----------|------------|------------|

- D14.1 Mount the NaCl crystal, TEL 582.004, in the crystal post (see Part I, para 10.9) ensuring that the major face having "flat matt" appearance is in the reflecting position (see Para D27.30).
- D14.2 Locate Primary Beam Collimator 582.001 in the Basic Port with the Imm slot vertical.
- D14.3 Mount Slide Collimator (3mm) 562.016 at E.S.13 and Collimator (1mm) 562.015 at E.S.18.
- D14.4 Zero-set and lock the Slave Plate and the Carriage Arm cursor as precisely as possible (see Part I, para. 10.6).
- D14.5 Sight through the collimating slits and observe that the primary beam direction lies in the surface of the crystal.

D14.6 Mount the G.M tube and it's holder at E.S.26.

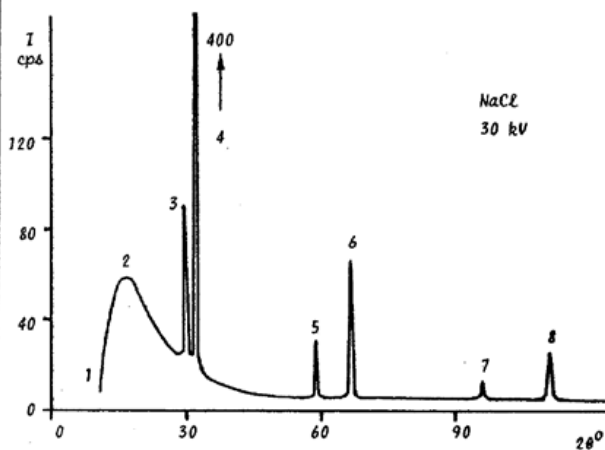
D14.7 Using a Ratemeter track the Carriage Arm round from it's minimum setting (about 11°, 2 $\theta$ ) to maximum setting (about 124°, 2 $\theta$ ).

Plot on graph paper the count rate per second at 1° (2 $\theta$ ) intervals, allowing time at each reading to estimate the mean of the fluctuations of the needle.

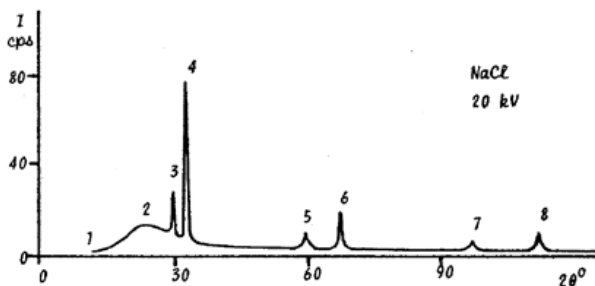
The Carriage Arm should be indexed to 15° (2 $\theta$ ) and the thumb wheel set to zero; when the Scatter Shield is closed, settings from 11° to 19° can be achieved using only thumb-wheel indications.

If the ratemeter has a loud speaker, note the random "quantum" nature of the beam of radiation at low count rates.

Where the count rate appears to peak, plot intervals of only 10' arc using the thumb-wheel (see Part I, para 10.8); at each peak, measure and record the maximum count rate and the angle  $2\theta$  as precisely as possible.



D14.8 Select 20kV and repeat D14.7.



Observe that the continuous spectra of "white" radiation exhibit peak intensities (feature 2) and intercepts on the  $2\theta$  axis (feature 1) which only vary with the voltage setting of the X-ray tube.

The six peaks, features 3 to 8, superimposed on the continuous spectrum do not vary in angle  $2\theta$  with voltage setting, but only in amplitude.

D14.9 Tabulate the results from the six superimposed peaks of the graph and calculate  $\lambda$  and  $n$ .

| Feature | $2\theta$ | $\theta$ | $\sin \theta$ | $2d$  | $n\lambda$ | $n$ |
|---------|-----------|----------|---------------|-------|------------|-----|
| 3       |           |          |               | 0.564 |            |     |
| 4       |           |          |               | 0.564 |            |     |
| 5       |           |          |               | 0.564 |            |     |

Observe that the sharp peaks are a pair of "emission lines" which re-appear in second and third orders of diffraction.

The more energetic radiation, termed K $\beta$ , is successively less intense than the longer wavelength, K $\alpha$ , line.

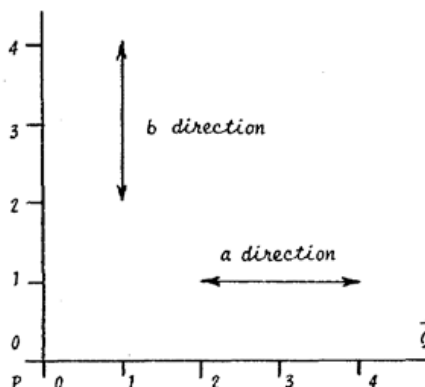
In the absence of micro-gratings, the "reasonable argument" was formulated by Sir Lawrence Bragg that the NaCl crystal could be used as a 3-dimensional grating which would reveal diffraction information by means of which the wavelength of the primary radiation could be established; but in the second sentence of D14 a bland assumption is made for Avogadro's number. If, however, the wavelength of the radiation is obtained by using a man-made grating, as in D13 then a contemporary approach is to reverse the sequence of the Bragg argument to provide an accurate evaluation of Avogadro's number.

Whichever didactic sequence is adopted, the Bragg experiment verifies that the incident radiation is both electromagnetic and heterogeneous and that co-operative interference can be induced using a crystal as a diffraction grating.

The crystal itself cannot be considered as the source of the 'dual' spectrum due to photon bombardment; the continuous spectrum is modified in both minimum wavelength and general intensity only by changing the X-ray tube accelerating voltage, without any variations in crystal parameters; the "emission lines" are particularly discreet in angle ( $2\theta$ ) whereas radiation from the crystal due to photon bombardment would be multi-directional.

The radiation must be derived through some "inverse photo-electric effect" from the impact of the thermionic electrons on the Copper target within the X-ray tube.

constructing at right angles to PQ; it is assumed that the Unit Cell is cubic and since  $a = b$  nm a cursory investigation shows that identical  $2\theta$  angles will be predicted.



D25.2 Set up for Bragg reflection as in D24.7, D24.8 and D24.9 and verify that the angle  $2\theta$  for diffraction peaks and "extinctions" are the same for the 'b' direction as for the 'a' direction.

The crystal has now been mapped in two of the three major apparent axes, 'a' and 'b'; sufficient information now exists to predict the reflections from planes perpendicular to other axes which are not major.

D25.3 Construct a lattice by projecting the perpendiculars to the integer points in both the 'a' and 'b' directions.

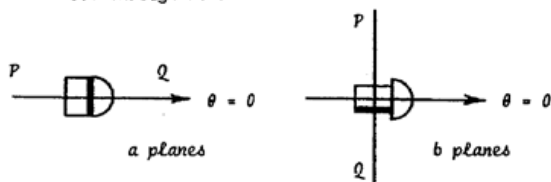
## D25 - THE UNIT CELL, TWO DIMENSIONS (40 MINUTES)

|             |       |            |            |
|-------------|-------|------------|------------|
| KIT 582+583 | 30 kV | 80 $\mu$ A | NORMAL LAB |
|-------------|-------|------------|------------|

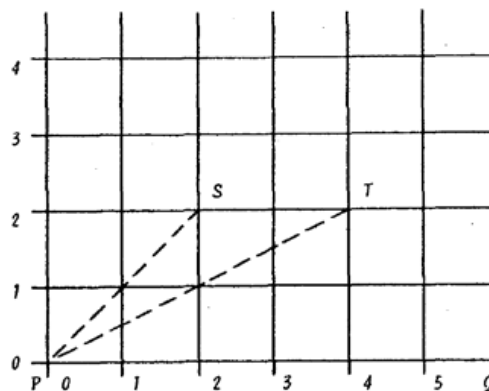
To confirm the hypothesis that the Unit Cell is cubic, the length of side must be established in the 'b' direction.

In order to achieve this the crystal must be oriented to ensure that the 'b' planes are parallel with the primary beam when  $2\theta = 0^\circ = \theta$ .

D25.1 Unlock the crystal jaw-clamp and rotate the crystal itself through  $90^\circ$ .



The 'b' planes can be drawn on the graph of D24.9 by constructing at right angles to PQ; it is assumed that the Unit Cell is cubic and since  $a = b$  nm a cursory investigation shows that identical  $2\theta$  angles will be predicted.



Any point on the lattice such as S now has co-ordinates (2,2) and T has (4,2).

These points represent PLANES within the crystal; since the graph is constructed from reciprocal dimensions it is called the Reciprocal Lattice and co-ordinates are usually referred to in terms of ( $hk$ ), the Miller indices.

In terms of two dimensions, the reference for point S is (22) where  $h = 2$  and  $k = 2$  and in real terms within the crystal the co-ordinates are  $(\frac{a}{2}, \frac{b}{2})$ .

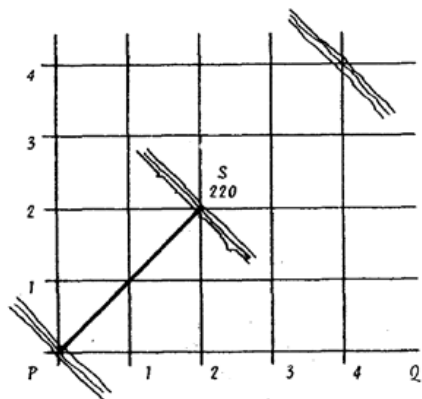
D25.4 In considering a three-dimensional structure the equivalent Miller index for the 'c' direction is the letter 'L'; the crystal has been located in the Crystal Post with the 'c' planes horizontal and parallel to the primary beam and with no additional orientation the index 'L' is zero.

The spatial co-ordinates for the S planes and T planes are thus (220) and (420) respectively.

D25.5 The spacing of the (220) planes is PS and therefore PS represents the direction of the normal to the planes; again this normal must be set perpendicular to the primary beam when  $2\theta = 0$ .

The k planes were oriented through  $90^\circ$  with respect to the h planes in D25.1; in this instance, the crystal must be oriented through  $SPQ$  when the carriage arm is set at  $2\theta = 0$ .

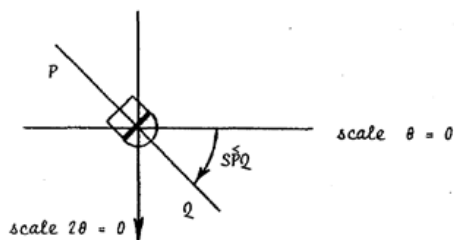
D25.6 Draw the line SP on the constructed diagram of D25.1 and, using the Calculator as a protractor, measure the angle  $SPQ$ .



D25.7 Set up for Bragg reflection as in D22.1 and D22.3.

D25.8 Unlock the 2:1 drive by turning the Knurled Clutch Plate in an anti-clockwise direction; with the

Carriage Arm accurately held at zero, rotate the crystal in a clockwise direction through the measured angle  $SPQ$  as precisely as possible and lock up the Knurled Clutch Plate.



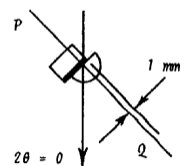
The normal to the (220) planes has been set perpendicular to the primary beam when the carriage arm is at zero.

D25.9 The normal to the (220) planes is also that for the (110), (330), (440), etc. and so  $2\theta$  angles can be read-off the constructed Reciprocal Lattice by using the Calculator; tabulate the measured angles.

| < SPQ | planes (hkl) | 2 θ         |        |
|-------|--------------|-------------|--------|
|       |              | theoretical | actual |
|       | 110          |             |        |
|       | 220          |             |        |
|       | 330          |             |        |
|       | 440          |             |        |
|       | 430          |             |        |
|       | 320          |             |        |
|       | 640          |             |        |
|       | 210          |             |        |
|       | 420          |             |        |
|       | 630          |             |        |
|       | 310          |             |        |
|       | 620          |             |        |

Observe that  $2\theta$  for the (110) planes is smaller than  $SPQ$  and considerable absorption of the primary beam might be expected.

D25.10 Unlock the jaw clamp and carefully displace the crystal such that the corner nearest the primary beam is about 1mm from the axial centre line scribed on the Crystal Post.



D25.11 Using a Ratemeter, search for the predicted diffraction peaks and tabulate the results, always noting any "extinctions", in the table of D25.9.

Observe that, as with Table D24.6, whenever  $h$  or  $k$  is odd there is an extinction.

D25.12 The length of PS, diagram D25.6, from simple trigonometry is

$$\sqrt{(2^2 + 2^2)}$$

or in three-dimensional terms

$$\sqrt{(h^2 + k^2 + l^2)}$$

In real terms within the crystal

$$d_{220} = \frac{a}{\sqrt{(2^2 + 2^2 + 0)}} \text{ nm}$$

and in general, for an isometric (cubic) system

$$d_{hkl} = \frac{a}{\sqrt{(h^2 + k^2 + l^2)}} \text{ nm}$$

Evaluate the 20 largest possible spacings of sets of planes within the Unit Cell by allocating suitable integers to  $h$ ,  $k$  and  $l$  such that  $h^2 + k^2 + l^2 = 1, 2$  et

| $h^2 + k^2 + l^2$ | $h$ | $k$ | $l$ | Reflections R<br>Extinctions X |
|-------------------|-----|-----|-----|--------------------------------|
| 1                 | 1   | 0   | 0   |                                |
| 2                 | 1   | 1   | 0   |                                |
| 3                 | 1   | 1   | 1   |                                |
| 4                 | 2   | 0   | 0   |                                |
| 5                 | 2   | 1   | 0   |                                |
| 6                 | 2   | 1   | 1   |                                |
| 7                 | —   | —   | —   | —                              |
| 8                 | 2   | 2   | 0   |                                |
| 9                 | 2   | 2   | 1   |                                |
|                   | 3   | 0   | 0   |                                |
| 10                | 3   | 1   | 0   |                                |

|    |   |   |   |   |
|----|---|---|---|---|
| 11 | 3 | 1 | 1 |   |
| 12 | 2 | 2 | 2 |   |
| 13 | 3 | 2 | 0 |   |
| 14 | 3 | 2 | 1 |   |
| 15 | — | — | — | — |
| 16 | 4 | 0 | 0 |   |
| 17 | 3 | 2 | 2 |   |
|    | 4 | 1 | 0 |   |
| 18 | 3 | 3 | 0 |   |
|    | 4 | 1 | 1 |   |
| 19 | 3 | 3 | 1 |   |
| 20 | 4 | 2 | 0 |   |

Observe that the seventh and fifteenth terms of the series are absent since 7 and 15 are not possible combinations of  $h^2 + k^2 + l^2$ .

D25.13 Compare the results with Table D24.6 for  $(h\ 0\ 0)$  planes and table D25.9 for  $(h\ k\ 0)$  planes, noting in the last column the observed presence or absence of reflections.

To verify the presence or absence of reflections from the 9 planes where  $l$  is not zero it is first necessary to establish that the "space-lattice" is indeed cubic by evaluating the length of the side of the Unit Cell in the 'c' direction.

The geometry of the Tel-X-Ometer will not allow the G. M. Tube to be rotated in a vertical direction; furthermore the construction of the Crystal Post does not allow the crystals to be conveniently oriented to present an adequate area to the primary beam.

An alternative technique can be used where a photographic record is made of reflections from a small rotating single crystal.